From the above activity, we get the result as, the sum of three angles of any triangle is 180°.

Now we prove the result in a formal method.

Given: A triangle ABC, where $\angle A = x$, $\angle B = y$ and $\angle C = z$.

Now let us prove, $x + y + z = 180^\circ$.

To show this we need to extend BC to D and draw a line CE, parallel to AB.

Now CE makes two angles, $\angle ACE$ and $\angle ECD$.

Let it be u and v respectively.

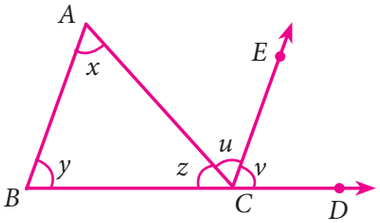


Fig. 4.4

Now z, u, v are the angles formed at a point on a straight line.
Therefore $z + u + v = 180^\circ$. ... (1)

Since AB and CE are parallel and DB is a transversal,

$v = y$ (corresponding angles).

Again AB and CE are parallel lines and AC is a transversal,

$u = x$ (alternate angles). Also $z + u + v = 180^\circ$ [by (1)]

Hence, by replacing u as x and v as y, we get $x + y + z = 180^\circ$.
Hence the sum of all three angles in a triangle is 180°.

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Example 4.1 Can the following angles form a triangle?

(i) 80°, 70°, 50° (ii) 56°, 64°, 60°

Solution

(i) Given angles 80°, 70°, 50°

Sum of the angles = 80°+70°+ 50° = 200° ≠ 180°

The given angles cannot form a triangle.

(ii) Given angles 56°, 64°, 60°

Sum of the angles = 56°+ 64°+ 60° = 180°

The given angles can form a triangle.

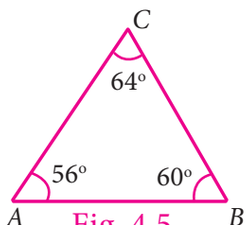


Fig. 4.5

Example 4.2 Find the measure of the missing angle in the given triangle ABC.

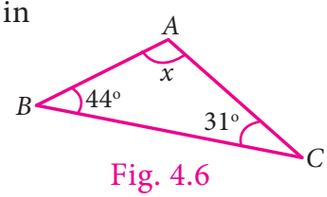


Fig. 4.6

Solution

Let $\angle A = x$

We know that, $\angle A + \angle B + \angle C = 180^\circ$ (angle sum property)

$x + 44^\circ + 31^\circ = 180^\circ$

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